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NOTES ON REPRESENTATION THEORY

Introduction

Representation theory is an immense topic. Most sources we know emphasize rigor and depth over conceptual clarity, especially for physics. Such presentations can conceal the regularity and beauty of the subject just due to their style and length. Here, we present the bare minimum, from the physicist's perspective, featuring rotation in three dimensions as a motivating example. We thus specifically bridge classical physics, where representation theory is not typically introduced, and quantum physics, where a student typically first encounters representation theory. We consider it unfortunate to delay the introduction of representation theory because it can be useful and clarifying in classical physics and because quantum physics already has many other steep hills to climb without representation theory. We deem it better to cover representation theory early in a curriculum, as a separate topic pertinent to both quantum and non-quantum physics.

Ontology

Object Inventory

Table 1: Inventory of Lie Theory Components

Short Name	Item Description	Context / Physical Interpretation
$\mathfrak{g}$	Abs $\mathfrak{su}(2)$	Abstract Lie algebra: $[E_i, E_j] = \epsilon_{ijk} E_k$
$\mathfrak{su}(2)_2$	$\mathfrak{su}(2)$ 2x2 Rep	Complex matrix realization (Generators T_k)
$\mathfrak{so}(3)_3$	$\mathfrak{so}(3)$ 3x3 Rep	Real matrix realization (Generators J_k)
$SU(2)$	$SU(2)$ Group	2×2 unitary matrices, $\det(=)1$
$SO(3)$	$SO(3)$ Group	3×3 orthogonal matrices, $\det(=)1$
V_{spin}	Spinors ($\mathbb{C}^2$)	Space for Fundamental Action of $SU(2)$
$\mathbb{R}^3$	Real $\mathbb{R}^3$	Euclidean space of physical 3-vectors
$\mathfrak{h}_\sigma$	Mat-Vecs	Traceless Hermitian 2×2 matrices
σ_k	Pauli Matrix	Basis $\{\sigma_1, \sigma_2, \sigma_3\}$ for the Pauli Map
f_{ijk}	Struct Consts	Numbers characterizing the bracket
$\mathbb{Z}$	Center	The kernel $\{I, -I\}$ of the $SU(2) \rightarrow SO(3)$ map
S^3	Topo S^3	Topology of $SU(2)$ (3-sphere)
$\mathbb{RP}^3$	Topo $\mathbb{RP}^3$	Topology of $SO(3)$ (Projective space)
ad_X	Infit. Ad	Adjoint action $[X, \cdot]$
Ad_Q	Finite Ad	Group action $Q \cdot Z \cdot Q^\dagger$
B	Killing Form	Trace inner product for coordinate extraction

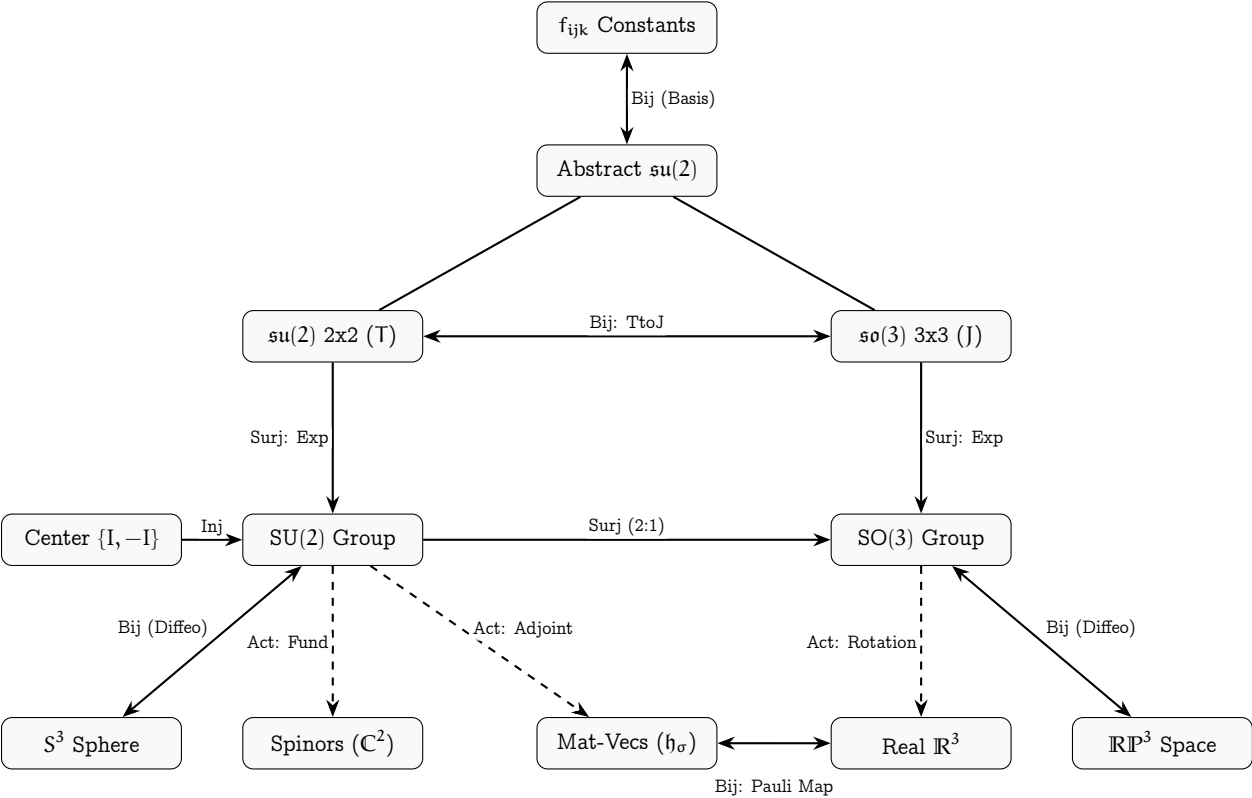
Mapping Inventory

Table 2: Registry of Mappings and Actions

Source Set	Target Set	Name / Mechanism	Type
$\mathfrak{su}(2)_2$	$SU(2)$	MatrixExp	Surj
$\mathfrak{so}(3)_3$	$SO(3)$	MatrixExp	Surj
$SU(2)$	$SO(3)$	Adjoint Rep / 2-to-1 Cover	Surj (2:1)
$\mathbb{Z}$	$SU(2)$	Center inclusion	Inj
$\mathbb{R}^3$	$\mathfrak{h}_\sigma$	Pauli Map ($\vec{v} \cdot \vec{\sigma}$)	Bij
$\mathfrak{h}_\sigma$	$\mathbb{R}^3$	Trace Projection (getTCoeffs)	Bij
$\mathfrak{su}(2)_2$	$\mathfrak{so}(3)_3$	Algebra Ad Rep (TtoJ)	Bij
$\mathfrak{g}$	f_{ijk}	Basis Mapping	Surj
$SU(2)$	S^3	Topo ID	Bij (Diffeo)
$SO(3)$	$\mathbb{RP}^3$	Topo ID	Bij (Diffeo)
$SU(2)$	$\mathbb{V}_{\text{spin}}$	Fund Act ($Q\psi$)	Act
$SU(2)$	$\mathfrak{h}_\sigma$	Ad Act ($QZQ^\dagger$)	Act
$SO(3)$	$\mathbb{R}^3$	Standard Rot ($R\vec{v}$)	Act

Relationships

Figure 1: Ontology of Objects and Mappings for Rotations in 3D



Adjoint and Fundamental Representations

The Adjoint Representation of $\mathfrak{su}(2)$

Consider two representations of the abstract Lie Algebra $\mathfrak{su}(2)$: the *adjoint representation* $\mathfrak{so}(3)$ and the *fundamental representation* $\mathfrak{su}(2)$, which happens to have the same name as the abstract algebra.

The **adjoint representation** $\mathfrak{so}(3)$ is spanned by the real **generators** J_k , defined by the specific structure constants $[J_i, J_j] = \epsilon_{ijk} J_k$. These structure constants model the *cross product*, which is the way of describing rotation in elementary physics. In the standard basis of the 9D vector space of 3×3 real matrices, the J_k are:

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Mathematica

```
J = {
  {{0, 0, 0}, {0, 0, -1}, {0, 1, 0}},
  {{0, 0, 1}, {0, 0, 0}, {-1, 0, 0}},
  {{0, -1, 0}, {1, 0, 0}, {0, 0, 0}}
};
```

These 3×3 , real, skew-symmetric matrices satisfy the same commutation relations as the members of the abstract algebra $\mathfrak{su}(2)$.

Cyclic Commutation

We verify that $[J_i, J_j] = \epsilon_{ijk} J_k$.

Mathematica

```
comm[M_][i_, j_] := M[[i]].M[[j]] - M[[j]].M[[i]];
commTest[M_][i_, j_, k_] := comm[M][i, j] == M[[k]];
triples = NestList[RotateLeft, Range[3], 2];
proof[And @@ commTest[J] @@@ triples]
```

The Fundamental Representation of $\mathfrak{su}(2)$

The fundamental representation is the most faithful. It is a 2×2 complex representation, spanned by the Pauli matrices. It is an isomorphism, unlike the adjoint representation. The adjoint representation maps two different members of $\mathfrak{su}(2)$ to one matrix.

Definition

The generators T_k are defined as $T_k = -\frac{i}{2}\sigma_k$. They also enjoy cyclic commutation.

```

----- Mathematica -----
T = (-I/2) PauliMatrix /@ {1, 2, 3};
proof[And @@ commTest[T] @@@ triples]

```

The Commutative Diagram Proof

The mapping $\pi : \mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$ preserves the exponential structure. This means that mapping an algebra element and then exponentiating is equivalent to exponentiating first and then mapping the group element.

```

----- Mathematica -----
(* Commutative Diagram Check *)
commutativeDiagram[X_] := FullSimplify[
  MatrixExp[ $\pi$ Map[X]] == ComplexExpand[ $\Pi$ map[MatrixExp[X]]]
];
proof[And @@ commutativeDiagram /@ ( $\theta$  T)]

```

The Big Beautiful Example: Adjoint Action

This final proof demonstrates that rotating a real vector $\vec{v} \in \mathbb{R}^3$ using an $SO(3)$ matrix R is identical to the Adjoint Action of $SU(2)$ on the corresponding matrix-vector $v_{\mathfrak{h}_\sigma} \in \mathfrak{h}_\sigma$.

Equivalence of Results

We build a rotation from Euler-like angles (α, β, γ) and then compare the two paths through the commutation diagram.

```

----- Mathematica -----
vR3 = {x, y, z};
{φ, θ, ψ} = {α, β, γ};

(* Path 1: \SO{3} Rotation *)
R = MatrixExp[φ J[[3]]] .
    MatrixExp[θ J[[2]]] .
    MatrixExp[ψ J[[1]]];
vPrimeR3 = Simplify[R . vR3];

(* Path 2: \SU{2} Adjoint Action *)
Q = MatrixExp[φ T[[3]]] .
    MatrixExp[θ T[[2]]] .
    MatrixExp[ψ T[[1]]];
vPrimeH0 = Simplify[ComplexExpand[
    Q . toH0[vR3] . ConjugateTranspose[Q]]];

(* The Bridge *)
proof[Simplify[fromH0[vPrimeH0] == vPrimeR3]]

```


Formal Proofs and Mathematica Verification

Adjoint Representation Algebra

The adjoint representation $\mathfrak{so}(3)$ is characterized by 3×3 real skew-symmetric matrices.

Generator Definition

The generators J_k are defined such that they satisfy the relation $[J_i, J_j] = \epsilon_{ijk} J_k$. In the standard basis:

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

```
Mathematica
J = {
  {{0, 0, 0}, {0, 0, -1}, {0, 1, 0}},
  {{0, 0, 1}, {0, 0, 0}, {-1, 0, 0}},
  {{0, -1, 0}, {1, 0, 0}, {0, 0, 0}}
};
```

Verification of the Lie Bracket

We verify the cyclic commutation relations $[J_1, J_2] = J_3$, $[J_2, J_3] = J_1$, and $[J_3, J_1] = J_2$.

```
Mathematica
comm[M_][i_, j_] := M[[i]].M[[j]] - M[[j]].M[[i]];
triples = NestList[RotateLeft, Range[3], 2];
And @@ (comm[J][#[[1]], #[[2]]] == J[[#[[3]]]] & /@ triples)
```

Fundamental Representation Algebra

The fundamental representation $\mathfrak{su}(2)$ is spanned by 2×2 complex traceless anti-Hermitian matrices.

Definition via Pauli Matrices

The generators T_k are proportional to the Pauli matrices σ_k :

$$T_k \equiv -\frac{i}{2} \sigma_k$$

Mathematica

```
T = (-I/2) PauliMatrix /@ {1, 2, 3};
```

Trace Inner Product and Metric

The Killing form on $\mathfrak{su}(2)$ is proportional to the trace inner product.

For the T basis, we have $\text{Tr}(T_i T_j) = -\frac{1}{2} \delta_{ij}$.

Mathematica

```
Table[Tr[T[[i]] . T[[j]]], {i, 3}, {j, 3}] // MatrixForm
(* Returns -1/2 IdentityMatrix[3] *)
```

The Adjoint Action Bridge

The core isomorphism between $SU(2)$ and $SO(3)$ is realized by the adjoint action of the group on its own algebra.

Mapping Matrices to Vectors

An element $X \in \mathfrak{su}(2)$ can be mapped to a vector $\vec{v} \in \mathbb{R}^3$ using the trace projection:

$$v_k = -2\text{Tr}(X \cdot T_k)$$

Mathematica

```
fromH0[mat_] := Table[-2 Tr[mat . T[[k]]], {k, 3}];
toH0[v_List] := Total[v * T];
```

Symbolic Proof of Rotational Equivalence

Let $Q \in SU(2)$ be a group element and $v \in \mathfrak{h}_\sigma$ be a matrix-vector. The adjoint action $QvQ^\dagger$ produces a new matrix-vector $v' \in \mathfrak{h}_\sigma$. We prove this is equivalent to the $SO(3)$ rotation $R\vec{v}$.

$$\text{to}_{\mathfrak{h}_\sigma}(R \cdot \vec{v}) \stackrel{!}{=} Q \cdot \text{to}_{\mathfrak{h}_\sigma}(\vec{v}) \cdot Q^\dagger$$

Mathematica

```
(* Parameters *)
vR3 = {x, y, z};
{φ, θ, ψ} = {α, β, γ};

(* Path A: \SO{3} *)
R = MatrixExp[φ J[[3]]] . MatrixExp[θ J[[2]]] . MatrixExp[ψ J[[1]]];
vPrimeSO3 = Simplify[R . vR3];
```

```

(* Path B: \SU{2} Adjoint *)
Q = MatrixExp[φ T[[3]]] . MatrixExp[θ T[[2]]] . MatrixExp[ψ T[[1]]];
vPrimeSU2 = Simplify[ComplexExpand[Q . toH0[vR3] . ConjugateTranspose[Q]]];

(* QED Verification *)
Simplify[fromH0[vPrimeSU2] == vPrimeS03]

```


Notation

We employ Schwichtenberg's¹ variations of the equality sign:

Symbol	Meaning
$\equiv$	Is defined as
$\triangleq$	Corresponds to / is represented by
$\stackrel{!}{=}$	Is required to be / must hold that

¹ Jakob Schwichtenberg. *Physics from Symmetry*. Undergraduate Lecture Notes in Physics. Springer International Publishing, Cham, Switzerland, 2nd edition, 2018. ISBN 978-3-319-88288-8. DOI: 10.1007/978-3-319-66631-0

Prerequisites

Arfken,² Spivak,³ and Baez and Muniain⁴ are good sources for background on *groups*, *vector spaces* and *differentiable manifolds*.

We adhere to the following terse definitions in the rest of these notes.

Definition 1 (Lie Algebra). A Lie algebra is a vector space $\mathfrak{g}$ equipped with a bilinear, anticommutative, binary operation $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ that satisfies the Jacobi identity:

$$[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$$

Definition 2 (Lie Group). A Lie group is a group and also a differentiable manifold.

Remark 1. A Lie algebra is not a group under the Lie bracket and is not a differentiable manifold.

² George B. Arfken, Hans J. Weber, and Frank E. Harris. *Mathematical Methods for Physicists*. Academic Press, Oxford, 7th edition, 2012. ISBN 9780123846549

³ Michael Spivak. *Calculus on Manifolds: A Modern Approach to Classical Theorems of Advanced Calculus*. W. A. Benjamin, Inc., New York, 1965. ISBN 9780805390216

⁴ John C. Baez and Javier P. Muniain. *Gauge Fields, Knots and Gravity*, volume 4 of *Series on Knots and Everything*. World Scientific, Singapore, 1994. ISBN 978-981-02-2034-1

Generators

The rest of this document concerns the relationships between Lie groups and Lie algebras. The first of these relationships comes via *generators*.

Definition 3 (Generator). A generator $X \in \mathfrak{g}$ is a Lie-algebra element that, multiplied by a real θ ,⁵ produces an element of a Lie group G under the (abstract) exponential map:

$$\begin{aligned}\exp : (\mathfrak{g} \times \mathbb{R}) &\rightarrow G \\ h(\theta) &= \exp(\theta X)\end{aligned}$$

Remark 2. A generator $X \in \mathfrak{g}$ in the Lie algebra can be recovered from a Lie-group element $h(\theta)$ by (abstract) differentiation:

$$X = \left. \frac{dh(\theta)}{d\theta} \right|_{\theta=0}$$

A representation is a concretization

The exponential map above is abstract, as is the differentiation operation to recover the Lie-algebra element. $\exp$ must be further defined for any concretization. For the sake of computation, physicists work almost exclusively with matrix concretizations. In such concretizations, $\exp$ is a matrix exponential, defined by Taylor series. From this point on, assume all concretizations are matrix concretizations.

Loosely, a *representation* is a matrix that stands for a Lie-algebra or Lie-group element. We formalize *representation* later, but for the purpose of illustrating generators as matrices, we must choose representations.

The same Lie algebra can generate many Lie Groups

We will present some general results and then illustrate them with two particular examples of interest in physics. Those two examples are representations of one Lie algebra $\mathfrak{su}(2)$ that generate two different Lie groups. In the first representation, the *adjoint representation*,^{6,7} ele-

⁵ X is a vector in the Lie Algebra $\mathfrak{g}$ considered as a vector space, and, by the rules of vector spaces, vectors can always be multiplied by scalars from the underlying field

⁶ symbolised by rep_{adj}

⁷ no relation to the adjoint operation on matrices, which is complex-conjugate transpose.

ments of the Lie algebra are represented by 3×3 real matrices. In the second representation, the *fundamental representation*,⁸ elements are represented by 2×2 complex matrices. We will show how to map from one to the other and back, that is, the *isomorphism* between these two representations. The meaning of that isomorphism is that the two representations actually represent the same Lie algebra. However, the two generated Lie groups are *not* equivalent, and interesting physics follow from that fact.

For the time being, take the words *adjoint* and *fundamental* at face value. We will elaborate them extensively as we go along.

Some general results

Proposition 1. *Any real matrix generators $J \in \mathfrak{g}$ of a special orthogonal group $SO(N)$ are traceless and antisymmetric.*

Remark 3. $SO(N)$ is a group of $N \times N$ invertible⁹ matrices with two more properties:

- *Special* means that the elements of $SO(N)$ have unit determinant.
- *Orthogonal* means that the transpose O^T of any element equals its matrix inverse O^{-1} .

Remark 4. A generator in the Lie algebra, if it is to generate an $N \times N$ real matrix in the Lie group by the matrix exponential, must itself be represented by an $N \times N$ real matrix.

Proof. The proof proceeds from implications of the definition of a matrix exponential.

- *Traceless:* *Special* implies that the N -dimensional generators are traceless:¹⁰

$$\begin{aligned} \det(O) &\stackrel{!}{=} 1 \implies \det(e^{\theta J}) = e^{\theta \text{Tr}(J)} = 1 \\ e^{\theta \text{Tr}(J)} &= 1 \implies \theta \text{Tr}(J) = 0 \implies \text{Tr}(J) = 0 \end{aligned} \quad (\text{Sch:3.60})$$

by *Jacobi's formula* linking $\det(\exp)$ and $\exp(\text{Tr})$.¹¹

- *Antisymmetric:* *Orthogonal* means that elements O of $SO(3)$ satisfy $O^T O = O O^T \stackrel{!}{=} 1$. Therefore:

$$\begin{aligned} O^T O = e^{\theta J^T} e^{\theta J} &\stackrel{!}{=} 1 \implies e^{\theta (J^T + J)} = 1 \\ &\implies J^T = -J \end{aligned} \quad (\text{Sch:3.59})$$

by the Baker-Campbell-Hausdorff formula¹² for matrix exponentials, because J^T and J commute. The *transpose* operation lifts into the exponent, as can be seen by a Taylor expansion, because θ is a scalar.

⁸ symbolized by rep_{fd}

⁹ All elements of any group are invertible, by definition of a group

¹⁰ Equation numbers with prefixes, such as (Sch:3.60), are taken from particular references that are meant to be obvious from the prefix.

¹¹ Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2nd edition, 2012. ISBN 978-0-521-83940-2

¹² Brian C. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, volume 222 of *Graduate Texts in Mathematics*. Springer, 2nd edition, 2015. ISBN 978-3-319-13466-6. DOI: 10.1007/978-3-319-13467-3

□

The Adjoint Representation is a particular real 3x3 representation

SO(3) is our controlling example: it is the Lie group of rotations in 3D Euclidean space, familiar to all physics students.

Proposition 1 pertains to any special orthogonal group SO(N).

How do we get members of SO(3) from generators in $\mathfrak{su}(2)$? First, we state¹³ that $\mathfrak{su}(2)$ is a 3-dimensional vector space, so we can choose a basis of three linearly independent generators $\{X_1, X_2, X_3\}$ to span $\mathfrak{su}(2)$.

SO(3) is also three dimensional, in the different sense that it's full of 3×3 matrices.

The following basis makes up the adjoint representation of $\mathfrak{su}(2)$ that generates SO(3). This set is not a guess; we'll show later how to derive it from the fundamental representation.¹⁴

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (\text{Sch:3.61})$$

The following Mathematica code checks that these generators in $\mathfrak{su}(2)$ satisfy the Lie-algebra commutation relations $[J_i, J_j] = \epsilon_{ijk} J_k$.¹⁵

```

Mathematica
J1 = {{0, 0, 0}, {0, 0, -1}, {0, 1, 0}};
J2 = {{0, 0, 1}, {0, 0, 0}, {-1, 0, 0}};
J3 = {{0, -1, 0}, {1, 0, 0}, {0, 0, 0}};

J1.J2 - J2.J1 == J3 (* True *)
J2.J3 - J3.J2 == J1 (* True *)
J3.J1 - J1.J3 == J2 (* True *)

```

Mathematica can further check that the matrix exponentials of these generators span SO(3) as a matrix group. The following should convince you if you are familiar with elementary rotation matrices:

```

Mathematica
MatrixExp[\[Theta] J1]
{{1, 0, 0},
 {0, Cos[\[Theta]], -Sin[\[Theta]]},
 {0, Sin[\[Theta]], Cos[\[Theta]]}}

```

The Fundamental Representation is a distinguished complex 2x2 representation

Fact 1. *There is exactly one simply connected Lie group corresponding to each Lie algebra.*

¹³ without proof, for now

¹⁴ The Lie bracket can “generate” the basis (in a different sense of the word *generate*), from just two of the J because $[J_1, J_2] = J_3$, etc.

¹⁵ ϵ_{ijk} is the totally antisymmetric Levi-Civita symbol that equals 1 when ijk is an even permutation of 123, -1 when ijk is an odd permutation, and 0 otherwise. This expression does *not* use the Einstein summation convention; it's just ordinary scalar-vector (as matrix) multiplication for any i, j , and k .

Proof. See Spivak.¹⁶ □

We don't need yet to understand "simply connected" to appreciate that the simply connected Lie group is the important and distinguished one. For our controlling example, this group is $SU(2)$, the special *unitary* group of 2×2 matrices.

Remark 5.

- *Special* means that the elements of $SU(2)$ have unit determinant, as before.
- *Unitary* means that the complex-conjugate transpose $U^\dagger$ of any element $U \in SU(2)$ equals its matrix inverse U^{-1} . *Unitary* for complex matrices is the analog of *orthogonal* for real matrices.

Proposition 2. *Complex 2×2 generators $T \in \mathfrak{su}(2)$ of the special unitary group $SU(2)$ are traceless and anti-Hermitian.*¹⁷

Proof. Proof of the traceless property goes just as above.

Unitary means that the elements U of $SU(2)$ satisfy $U^\dagger U = U U^\dagger \stackrel{!}{=} 1$,¹⁸

Hermitian means that a matrix is self-adjoint. *Anti-Hermitian* means that a matrix equals the negative of its adjoint. Now we see that

$$U^\dagger U = e^{\theta^\dagger T} e^{\theta T} \stackrel{!}{=} 1 \implies T^\dagger \stackrel{!}{=} -T \quad (1)$$

The adjoint operation also lifts into the exponent, like *transpose*, because θ is a *real* scalar. □

A basis for this fundamental representation is $-i/2$ times the three *Pauli matrices*:¹⁹

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{Sch:3.80})$$

The basis is $T_1 = -i\sigma_1/2$, $T_2 = -i\sigma_2/2$, $T_3 = -i\sigma_3/2$ as can be readily verified in Mathematica, showing, again in passing, that the factor of $1/2$ maintains the unit *structure constants* ϵ_{ijk} . Without the $1/2$, we'd be chasing around factors of 2.

Mathematica

```
T1 = -I/2 PauliMatrix[1];
T2 = -I/2 PauliMatrix[2];
T3 = -I/2 PauliMatrix[3];
T1.T2 - T2.T1 == T3 (* True *)
T2.T3 - T3.T2 == T1 (* True *)
T3.T1 - T1.T3 == T2 (* True *)
```

¹⁶ Michael Spivak. *A Comprehensive Introduction to Differential Geometry*. Publish or Perish, Inc., Houston, Texas, 3rd edition, 1999. ISBN 9780914098706. Five Volumes

¹⁷ *Anti-Hermitian* for complex matrices is the analog of antisymmetric for real matrices.

¹⁸ As noted, this meaning of *adjoint* is distinct from its meaning in *adjoint representation*, which pertains to the representation of $SO(3)$.

¹⁹ The Pauli matrices are also not a guess. They are the unique 2×2 traceless Hermitian matrices in the complex domain. The factor of $-i$ makes them anti-Hermitian, and the factor of $1/2$ is convenient.

Representations

Representations are maps from a Lie group or a Lie Algebra to a vector space of matrices. The representations of a Lie group preserve the group properties and the representations of a Lie algebra preserve the Lie bracket. Formally:

Definition 4 (Lie Group Representation). A representation of a Lie group G is a map Π from any Lie group element $g \in G$ to a linear transformation $\Pi(g)$ of some vector space V such that group properties are preserved.

$$g \xrightarrow[\Pi]{} \Pi(g) \quad (\text{Sch:3.65})$$

Remark 6. Preserving group properties means:

Multiplication is preserved: $\Pi(g_1 g_2) = \Pi(g_1) \Pi(g_2)$

Group identity maps to matrix identity: $\Pi(e) = I$

Inverses are preserved: $\Pi(g^{-1}) = \Pi(g)^{-1}$

Definition 5 (Lie Algebra Representation). A representation of a Lie algebra $\mathfrak{g}$ is a map π from any Lie algebra element $X \in \mathfrak{g}$ to a linear transformation $\pi(X)$ of some vector space V such that the Lie bracket is preserved:

$$\pi([X, Y]) = [\pi(X), \pi(Y)] \quad (2)$$

where the bracket on the right is the standard matrix commutator:

$$[\pi(X), \pi(Y)] \equiv \pi(X)\pi(Y) - \pi(Y)\pi(X).$$

When a Lie group and a Lie algebra are linked by the exponential map, every representation of a group “lifts” to a representation of its algebra, and vice versa. Figure 2 shows the relationships amongst:

The Lie algebra $\mathfrak{g}$: e.g, $\mathfrak{su}(2)$, spanned by T in the fundamental and by J in the adjoint representation

The Lie group G : e.g., $G_{\text{sc}} = \text{SU}(2)$, the unique simply connected group, or $G = \text{SO}(3)$, the multiply connected group of rotation matrices in $\mathbb{R}^3$

The Carrier Space V : e.g., for $\text{SU}(2)$, $V = \mathbb{C}^2$, the *spinor space*; for $\text{SO}(3)$, $V = \mathbb{R}^3$, the physical Euclidean space

The General Linear Group $GL(V)$: the group of all invertible linear transformations of the carrier space

The General Linear Algebra $gl(V)$: the space of all linear maps from V to V that preserve the Lie bracket as the matrix commutator
 $[A, B] = AB - BA$

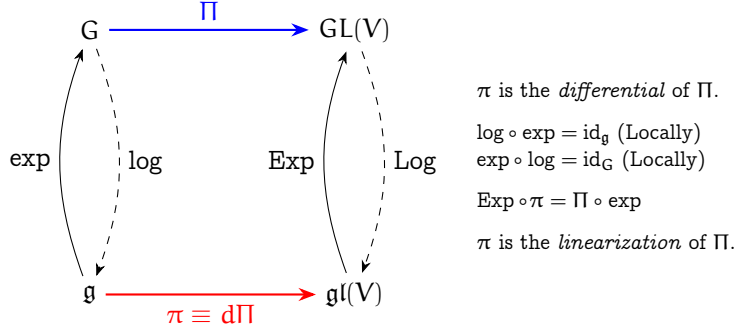


Figure 2: The relationship amongst Lie group and Lie algebra representations.

The Two Worlds of $su(2)$

In our motivating example of $su(2)$, we found two distinct bases: a 2×2 complex basis $\{T_a\}$ and a 3×3 real basis $\{J_a\}$. To a physicist, these are not just different sizes of matrices; they represent two different ways a group can act on the physical world.

The Fundamental Pair (Direct Action)

The 2×2 representation is the *fundamental representation*. It describes how the group $SU(2)$ acts on its “native” carrier space $V = \mathbb{C}^2$ (the space of spinors).

- **Group Level:** $\Pi_{\text{fund}}(Q)$ is the 2×2 matrix itself.
- **Algebra Level:** $\pi_{\text{fund}}(X)$ is the 2×2 generator.
- **Action:** The *direct action* $Q|\xi\rangle$, where $Q \in SU(2)$ and $|\xi\rangle$ is a spinor, produces a new spinor.

The Adjoint Pair (Adjoint Action)

The 3×3 representation is the *adjoint representation*. It describes how the group $SU(2)$ acts on its own Lie algebra $su(2)$.

- **Group Level:** $\Pi_{\text{adj}}(Q)$ is the 3×3 rotation matrix in $SO(3)$.
- **Algebra Level:** $\pi_{\text{adj}}(X)$ is the 3×3 matrix of structure constants
 $(J_a)_{bc} = \epsilon_{abc}$.

- **Action:** The *adjoint action* $QXQ^\dagger$, where $Q \in \text{SU}(2)$ and X is a vector in the algebra, produces a new vector in the algebra.

Remark 7. Note the “squaring” relationship: The adjoint representation is (roughly) the product of two fundamental representations ($Q \dots Q^\dagger$). This is why $\text{SO}(3)$ is the adjoint group of $\text{SU}(2)$. It is also why spinors ($\text{SU}(2)$) are the “square root” of vectors ($\text{SO}(3)$), and why it takes a 4π rotation to return a spinor to its original state, while 2π suffices for a vector. Another view is that the relationship between the fundamental and adjoint representations explains the *double covering* of $\text{SO}(3)$ by $\text{SU}(2)$.

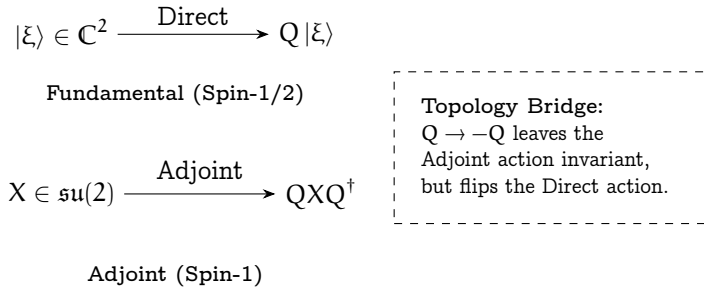


Figure 3: Direct vs. Adjoint actions of $\text{SU}(2)$.

Fact 2. *The Adjoint representation is always available for any Lie group. It is the “regulation” way to view a Lie group as a matrix group of rotations on its own algebra.*

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